

Suman Kumar

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Professional Summary

ECE undergraduate at BIT Sindri, passionate about staying ahead of emerging tech-agentic AI, RAG, LangChain, LangGraph, MCP. Built a C++ vector database from scratch (HNSW, KD-Tree, REST API) and led platforms serving 500+ members. Eager to bring that curiosity and first-principles rigor to an AI-native engineering role.

Education

BIT Sindri — B.Tech in Electronics & Communication Engineering Jul 2023 – 2027

CGPA: **7.8 / 10.0** | Dhanbad, Jharkhand

- Relevant Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, Digital Electronics, Communication Systems

Kendriya Vidyalaya — Class XII (Intermediate) 2023

Score: **89.4%** | Chandrapura

Experience

ONGC (Oil and Natural Gas Corporation) Winter 2025

Communication & Process Monitoring Trainee | Agartala, India

- Analyzed SCADA/PLC-based industrial protocols in a live production environment, monitoring link stability and ensuring data integrity across remote sensing units.
- Documented system-workflow specifications and protocol reports, enhancing cross-team technical communication skills in a regulated industry setting.

Ministry of Electronics & IT (*in collaboration with IIT ISM*) 2024

UAS Bootcamp Trainee | Dhanbad, India

- Led hardware integration in a 4-member team to assemble, calibrate, and deploy a Hexacopter UAS; executed autonomous flight missions with real-time sensor telemetry under a government-sponsored program (MeitY × IIT ISM).

Key Projects

AstroVector Engine [GitHub](#) | [Live Demo](#)

C++, REST API, Ollama, Llama 3.2, PCA, HNSW, KD-Tree

- Systems Design:** Architected a high-performance vector database from scratch using HNSW and KD-Tree indexing, achieving sub-millisecond semantic search at scale.
- AI Integration:** Engineered an offline RAG pipeline with a custom REST API and a locally-hosted LLM (Llama 3.2 via Ollama); served a real-time 2D PCA visualization dashboard — demonstrating full-stack, end-to-end ownership.

Official GRS Club Website — BIT Sindri

PHP, MySQL, JavaScript, HTML5 / CSS3

- Backend:** Designed a normalized MySQL schema for 500+ member registrations; automated event-data pipelines, cutting manual data-entry workload by **40%**.
- Frontend:** Delivered a responsive UI featuring event galleries and member portfolios; enforced code-review and QA workflows across the team.

Technical Skills

Languages C++, Python, JavaScript, PHP, SQL (MySQL), HTML5, CSS3

Tools Git, GitHub, VS Code

CS Concepts Data Structures & Algorithms (300+ problems), OOP, Vector Databases, RAG Pipelines,

Certifications Machine Learning Specialization - DeepLearning.AI & Stanford University (*6 months*)

Achievements & Leadership

- Competitive Programming:** Solved **300+ problems** on LeetCode and GeeksforGeeks, covering a broad range of DSA topics and paradigms.
- Club Leadership:** Placement Coordinator | Secretary and Technical Head **GRS Club, BIT Sindri** (500+ members); organized multiple college-wide technical and cultural events.
- Hackathons:** Runner-up (college Hackathon); 2nd Runner-up (PLACE-COM Mock GD).
- Sports:** National Basketball (1×); Regional (3×).